Listing 6.4 The adjacency matrix function

```
function adjacency() {
  queue()
  .defer(d3.csv, "nodelist.csv")
  .defer(d3.csv, "edgelist.csv")
  .await(function(error, file1, file2) {
          createAdjacencyMatrix(file1, file2);
  });

  function createAdjacencyMatrix(nodes,edges) {
    var edgeHash = {};
    for (x in edges) {
      var id = edges[x].source + "-" + edges[x].target;
      edgeHash[id] = edges[x];
    };
    matrix = [];
    for (a in nodes) {
      for (b in nodes) {
        var grid =
          {id: nodes[a].id + "-" + nodes[b].id,
               x: b, y: a, weight: 0};
        if (edgeHash[grid.id]) {
          grid.weight = edgeHash[grid.id].weight;
        };
        matrix.push(grid);
      };
    };

    d3.select("svg")
      .append("g")
      .attr("transform", "translate(50,50)")
      .attr("id", "adjacencyG")
      .selectAll("rect")
      .data(matrix)
      .enter()
      .append("rect")
      .attr("class", "grid")
      .attr("width", 25)
      .attr("height", 25)
      .attr("x", function (d) {return d.x * 25})
      .attr("y", function (d) {return d.y * 25})
      .style("fill-opacity", function (d) {return d.weight * .2;})

    var scaleSize = nodes.length * 25;
    var nameScale = d3.scale.ordinal()
         .domain(nodes.map(function (el) {return el.id}))
         .rangePoints([0,scaleSize],1);

    var xAxis = d3.svg.axis()
           .scale(nameScale).orient("top").tickSize(4);
    var yAxis = d3.svg.axis()
           .scale(nameScale).orient("left").tickSize(4);
    d3.select("#adjacencyG").append("g").call(yAxis);
    d3.select("#adjacencyG").append("g").call(xAxis)
           .selectAll("text")
```

We need to load two datasets before we can get started, and queue lets us move the asynchronous loaders into a synchronous format.

A hash allows us to test if a source-target pair has a link.

Creates all possible source-target connections

Sets the xy coordinates based on the source-target array positions

If there's a corresponding edge in our edge list, give it that weight.

Creates an ordinal scale from the node IDs

Used for ordinal values

Both axes use the same scale.